> # ***Лабораторная работа 3.2 "Обыкновенные дифференциальные уравнения высших порядков "***

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 # Вариант 2.

Задание №1.1

$$DY : x = y'' + \ln(y'')$$

$$(\ y''\ =\ t\ )$$

$$\begin{cases} y'' = t \\ x = t + \ln(t) \end{cases}$$

$$dx = \frac{t+1}{t}dt \text{ - нашли дифференциал}$$

$$d(y') = y'' \cdot dx = \left(t \cdot \frac{t+1}{t}\right)dt = (t+1)dt$$

$$y' = \frac{t^2}{2} + t + C$$

$$dy = \left(\frac{t^2}{2} + t + C\right)\left(\frac{t+1}{t}\right)dt$$

Получили ДУ с *разделяющимися переменными* и *нашли у*

```
> DY1 := t + ln(t);
DefPoPeremennouT_DY1 := diff(DY1, t);

MainDY := diff(y(t), t) = (t^2/2 + t + C) · ((t+1)/t);
ReshenieDY1 := dsolve(MainDY);
Graphics_ReshenieDY1 := rhs(%);
```

$$DY1 := t + \ln(t)$$

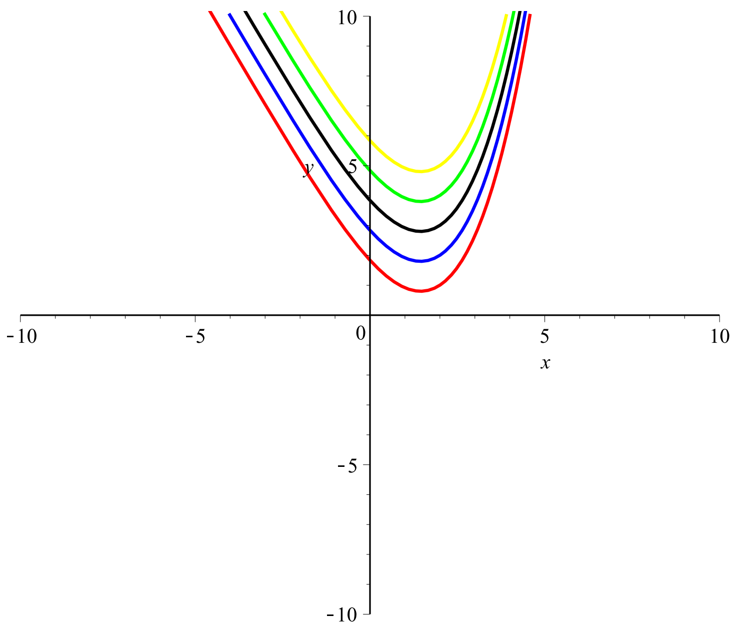
$$DefPoPeremennouT\_DY1 := 1 + \frac{1}{t}$$

$$MainDY := \frac{d}{dt}\, y(t) = \frac{\left(\frac{1}{2}\, t^2 + t + C\right)(t+1)}{t}$$

$$ReshenieDY1 := y(t) = \frac{1}{6}\, t^3 + C\, t + \frac{3}{4}\, t^2 + t + C\ln(t) + \_C1 \qquad (1)$$

$$Graphics\_ReshenieDY1 := \frac{1}{6}\, t^3 + C\, t + \frac{3}{4}\, t^2 + t + C \ln(t) + \_C1 \tag{1}$$

```
> Line1 := plot([DY1(t), subs(C=-2, _C1=1, Graphics_ReshenieDY1(t)), t=-10..10], x=-10
      ..10, y=-10..10, thickness=2, color=red) :
  Line2 := plot([DY1(t), subs(C=-2, _C1=2, Graphics_ReshenieDY1(t)), t=-10..10], x=-10
      ..10, y=-10..10, thickness=2, color=blue) :
  Line3 := plot([DY1(t), subs(C=-2, _C1=3, Graphics_ReshenieDY1(t)), t=-10..10], x=-10
      ..10, y=-10..10, thickness=2, color=black) :
  Line4 := plot([DY1(t), subs(C=-2, _C1=4, Graphics_ReshenieDY1(t)), t=-10..10], x=-10
      ..10, y=-10..10, thickness=2, color=green) :
  Line5 := plot([DY1(t), subs(C=-2, _C1=5, Graphics_ReshenieDY1(t)), t=-10..10], x=-10
      ..10, y=-10..10, thickness=2, color=yellow) :

  plots[display](Line1, Line2, Line3, Line4, Line5);
```




```
> restart
```

Задание №1.2

DY : (x^2 + 1)(yy'' - (y')^2) = 2xyy'

(U = $\frac{y'}{y}$) ,

y' = yU, y'' = y'U + U'y = yU^2 + U'y

:

U'(x^2 + 1) = 2xU

U = (x^2 + 1) * C

y U , :

$\frac{y'}{y}$ = (x^2 + 1)*C

: y = C 1 * $e^{C\left(\frac{x}{3}^{3}+x\right)}$

\> $DY2 := \frac{diff(U(x), x)}{U(x)} = \frac{2x}{(x^2+1)};$
$ReshenieDY2 := dsolve(DY2);$
$ReshenieDY2 := rhs(\%);$

$MainDY := \frac{diff(y(x), x)}{y(x)} = ReshenieDY2;$
$ReshenieMainDY := dsolve(MainDY);$
$ReshenieMainDY := rhs(\%);$

$$DY2 := \frac{\frac{d}{dx}U(x)}{U(x)} = \frac{2x}{x^2+1}$$

$$ReshenieDY2 := U(x) = \_C1\, x^2 + \_C1$$

$$ReshenieDY2 := \_C1\, x^2 + \_C1$$

$$MainDY := \frac{\frac{d}{dx}y(x)}{y(x)} = \_C1\, x^2 + \_C1$$

$$ReshenieMainDY := y(x) = \_C2\, e^{\frac{1}{3}x^3\,\_C1}\, e^{\_C1\, x}$$

$$ReshenieMainDY := \_C2\, e^{\frac{1}{3}x^3\,\_C1}\, e^{\_C1\, x} \tag{2}$$

\> $Line1 := plot(subs(\_C2 = 2, \_C1 = -5, ReshenieMainDY(x)), x = -10..10, y = -10..10, thickness$

$= 2$, $color = red$) :

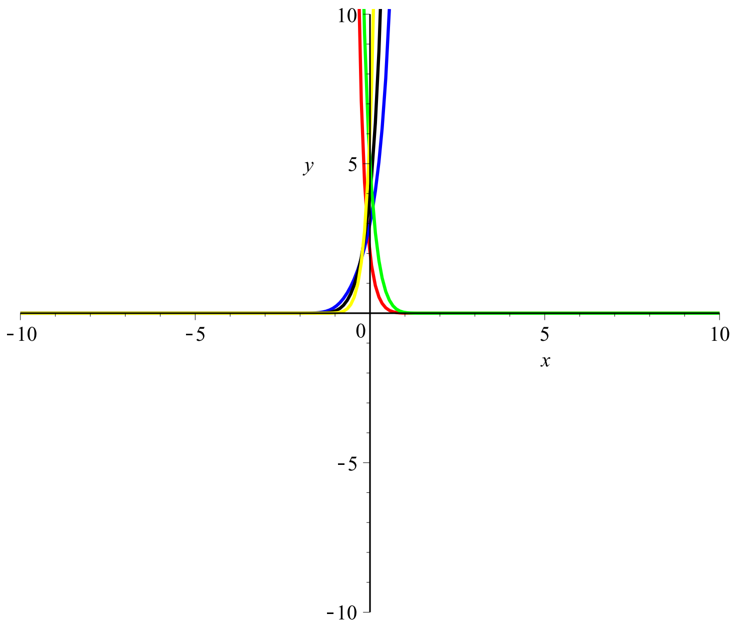
$Line2 := plot(subs(\_C2 = 3, \_C1 = 2, ReshenieMainDY(x)), x = -10 ..10, y = -10 ..10, thickness = 2, color = blue)$:

$Line3 := plot(subs(\_C2 = 4, \_C1 = 3, ReshenieMainDY(x)), x = -10 ..10, y = -10 ..10, thickness = 2, color = black)$:

$Line4 := plot(subs(\_C2 = 5, \_C1 = -4, ReshenieMainDY(x)), x = -10 ..10, y = -10 ..10, thickness = 2, color = green)$:

$Line5 := plot(subs(\_C2 = 6, \_C1 = 5, ReshenieMainDY(x)), x = -10 ..10, y = -10 ..10, thickness = 2, color = yellow)$:

$plots[display](Line1, Line2, Line3, Line4, Line5)$;



> $restart$

Задание №1.3

DY : y' = xy'' - e^(y'')

(U = y') ,

U = U'x - e^(U')

p = U '

dU = pdx

U = px - e^p

d U :

dU = pdx + xdp - e^p*dp

=> p = C ;

U = Cx - e^C

U = y' , y' = C x - e ^ C , :

$$y = \frac{Cx^2}{2} + e\text{^}C * x + C1$$

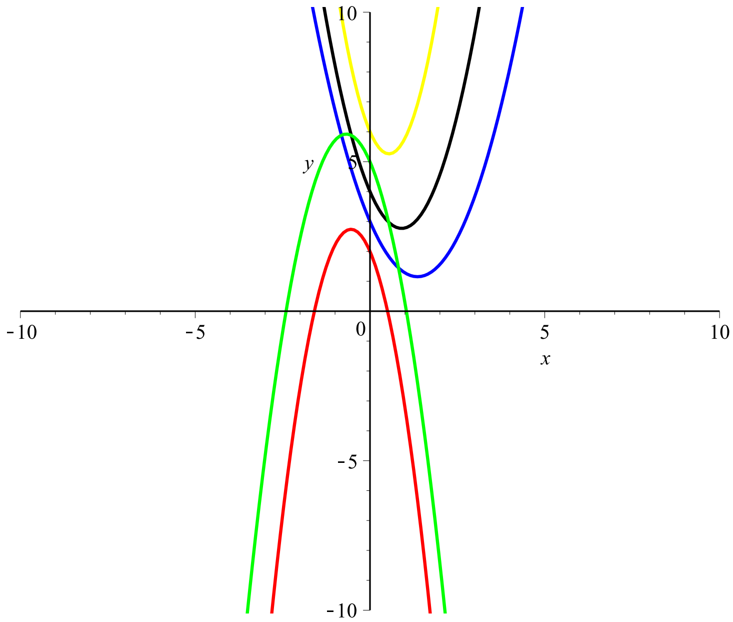
```
> DY3 := diff(y(x), x) = x·diff(y(x), x$2) - e(diff(y(x), x$2));

  ReshenieDY3 := dsolve(DY3);
  ReshenieDY3 := rhs(%);
```

$$DY3 := \frac{d}{dx}\, y(x) = x \left(\frac{d^2}{dx^2}\, y(x) \right) - e$$

$$ReshenieDY3 := y(x) = \frac{1}{2}\, \_C1\, x^2 - e\, x + \_C2$$

$$ReshenieDY3 := \frac{1}{2}\, \_C1\, x^2 - e\, x + \_C2 \qquad (3)$$

```
> Line1 := plot(subs(_C2 = 2, _C1 = -5, ReshenieDY3(x)), x = -10..10, y = -10..10, thickness
      = 2, color = red) :
  Line2 := plot(subs(_C2 = 3, _C1 = 2, ReshenieDY3(x)), x = -10..10, y = -10..10, thickness = 2,
      color = blue) :
  Line3 := plot( subs(_C2 = 4, _C1 = 3, ReshenieDY3(x)), x = -10..10, y = -10..10, thickness = 2,
      color = black) :
  Line4 := plot( subs(_C2 = 5, _C1 = -4, ReshenieDY3(x)), x = -10..10, y = -10..10, thickness
      = 2, color = green) :
  Line5 := plot( subs(_C2 = 6, _C1 = 5, ReshenieDY3(x)), x = -10..10, y = -10..10, thickness = 2,
      color = yellow) :
  plots[display](Line1, Line2, Line3, Line4, Line5);
```



> *restart*

Задание №1.4

: y'' = 2*($\frac{y'}{x}$ - $\frac{y}{x^2}$) + e^(1/x)/x^2

> $DY4 := diff(y(x), x\$2) = 2\cdot\left(\frac{diff(y(x), x)}{x} - \frac{y(x)}{x^2}\right) + \frac{\exp\left(\frac{1}{x}\right)}{x^2};$

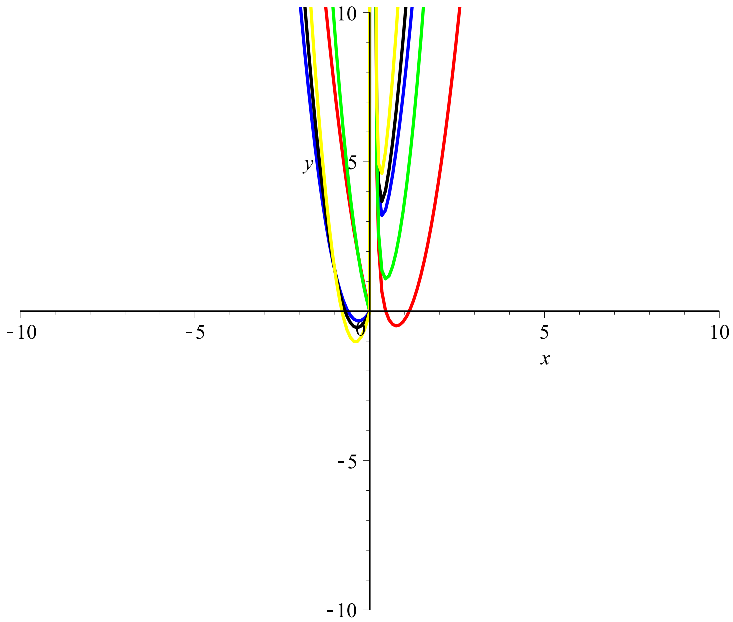
$ReshenieDY4 := dsolve(DY4);$
$ReshenieDY4 := rhs(\%);$

$$DY4 := \frac{d^2}{dx^2}\, y(x) = \frac{2\left(\frac{d}{dx}\, y(x)\right)}{x} - \frac{2\, y(x)}{x^2} + \frac{e^{\frac{1}{x}}}{x^2}$$

$$ReshenieDY4 := y(x) = x^2\, e^{\frac{1}{x}} + x^2\, \_C2 + x\, \_C1$$

$$ReshenieDY4 := x^2\, e^{\frac{1}{x}} + x^2\, \_C2 + x\, \_C1 \qquad (4)$$

> $Line1 := plot(subs(\_C2 = 2, \_C1 = -5, ReshenieDY4), x = -10..10, y = -10..10, thickness = 2, color = red) :$
$Line2 := plot(subs(\_C2 = 3, \_C1 = 2, ReshenieDY4(x)), x = -10..10, y = -10..10, thickness = 2, color = blue) :$
$Line3 := plot(subs(\_C2 = 4, \_C1 = 3, ReshenieDY4(x)), x = -10..10, y = -10..10, thickness = 2, color = black) :$
$Line4 := plot(subs(\_C2 = 5, \_C1 = -4, ReshenieDY4(x)), x = -10..10, y = -10..10, thickness = 2, color = green) :$
$Line5 := plot(subs(\_C2 = 6, \_C1 = 5, ReshenieDY4(x)), x = -10..10, y = -10..10, thickness = 2, color = yellow) :$
$plots[display](Line1, Line2, Line3, Line4, Line5);$



> *restart*

Задание №2

DY: xy''' + y'' = 1
(xy'')' = xy''' + y''
d(xy'' - x) = 0
xy'' - x = C
y'' = 1 + C/x (*)

(*) y .

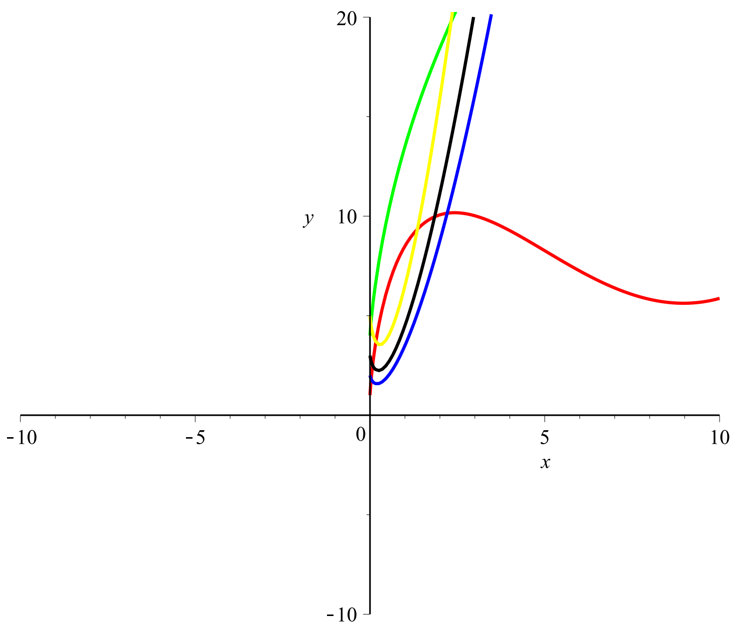
```
> DY5 := x·diff(y(x), x$3) + diff(y(x), x$2) = 1;

ReshenieDY5 := dsolve(DY5);
ReshenieDY5 := rhs(%);
```

$$DY5 := x\left(\frac{d^3}{dx^3}\,y(x)\right) + \frac{d^2}{dx^2}\,y(x) = 1$$

$$ReshenieDY5 := y(x) = \ln(x)\,x\,\_C1 - x\,\_C1 + \frac{1}{2}\,x^2 + \_C2\,x + \_C3$$

$$ReshenieDY5 := \ln(x)\,x\,\_C1 - x\,\_C1 + \frac{1}{2}\,x^2 + \_C2\,x + \_C3 \qquad \textbf{(5)}$$

```
> Line1 := plot(subs(_C2 = 2, _C1 = -5, _C3 = 1, ReshenieDY5(x)), x = -10..10, y = -10..20,
      thickness = 2, color = red) :
  Line2 := plot(subs(_C2 = 3, _C1 = 2, _C3 = 2, ReshenieDY5(x)), x = -10..10, y = -10..20,
      thickness = 2, color = blue) :
  Line3 := plot( subs(_C2 = 4, _C1 = 3, _C3 = 3, ReshenieDY5(x)), x = -10..10, y = -10..20,
      thickness = 2, color = black) :
  Line4 := plot( subs(_C2 = 5, _C1 = -4, _C3 = 4, ReshenieDY5(x)), x = -10..10, y = -10..20,
      thickness = 2, color = green) :
  Line5 := plot( subs(_C2 = 6, _C1 = 5, _C3 = 5, ReshenieDY5(x)), x = -10..10, y = -10..20,
      thickness = 2, color = yellow) :
  plots[display](Line1, Line2, Line3, Line4, Line5);
```



> *restart*

Задание №3

DY: y'' - 4y' + 4y = -e^2x * sin(6x)

:

f(x) = e^(alpha * x) * (Pm(x)cos(beta * x) + Qm(x)sin(beta * x))

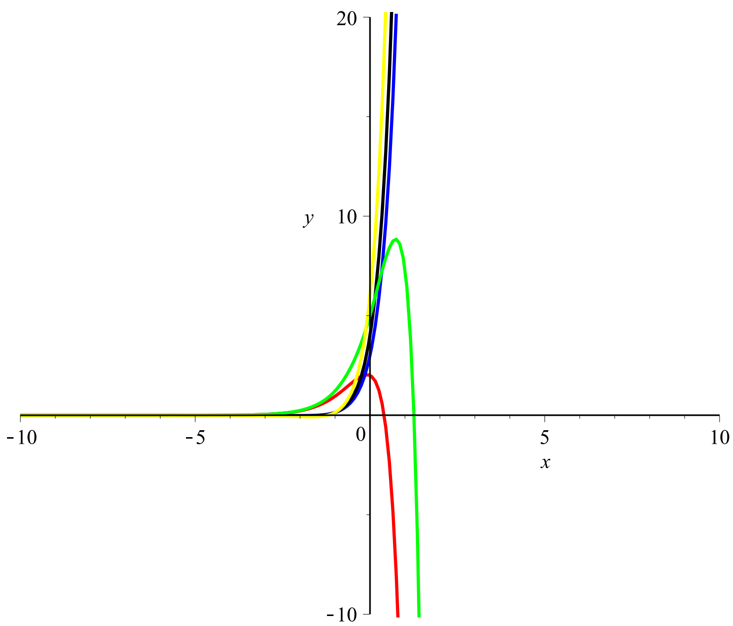
```
> DY6 := diff(y(x), x$2) - 4·diff(y(x), x) + 4·y(x) = -exp(2·x)·sin(6·x);

ReshenieDY6 := dsolve(DY6);
ReshenieDY6 := rhs(%);
```

$$DY6 := \frac{d^2}{dx^2} y(x) - 4 \left(\frac{d}{dx} y(x)\right) + 4\, y(x) = -e^{2x} \sin(6x)$$

$$ReshenieDY6 := y(x) = e^{2x}\, \_C2 + e^{2x} x\, \_C1 + \frac{1}{36} e^{2x} \sin(6x)$$

$$ReshenieDY6 := e^{2x}\, \_C2 + e^{2x} x\, \_C1 + \frac{1}{36} e^{2x} \sin(6x) \qquad \textbf{(6)}$$

```
> Line1 := plot(subs(_C2 = 2, _C1 = -5, _C3 = 1, ReshenieDY6(x)), x = -10..10, y = -10..20,
      thickness = 2, color = red) :
  Line2 := plot(subs(_C2 = 3, _C1 = 2, _C3 = 2, ReshenieDY6(x)), x = -10..10, y = -10..20,
      thickness = 2, color = blue) :
  Line3 := plot( subs(_C2 = 4, _C1 = 3, _C3 = 3, ReshenieDY6(x)), x = -10..10, y = -10..20,
      thickness = 2, color = black) :
  Line4 := plot( subs(_C2 = 5, _C1 = -4, _C3 = 4, ReshenieDY6(x)), x = -10..10, y = -10..20,
      thickness = 2, color = green) :
  Line5 := plot( subs(_C2 = 6, _C1 = 5, _C3 = 5, ReshenieDY6(x)), x = -10..10, y = -10..20,
      thickness = 2, color = yellow) :
  plots[display](Line1, Line2, Line3, Line4, Line5);
```


20
10
-10
y
-10
-5
0
5
10
x

>